



# L3 Review Questions

# Q1: Smoothing and localization

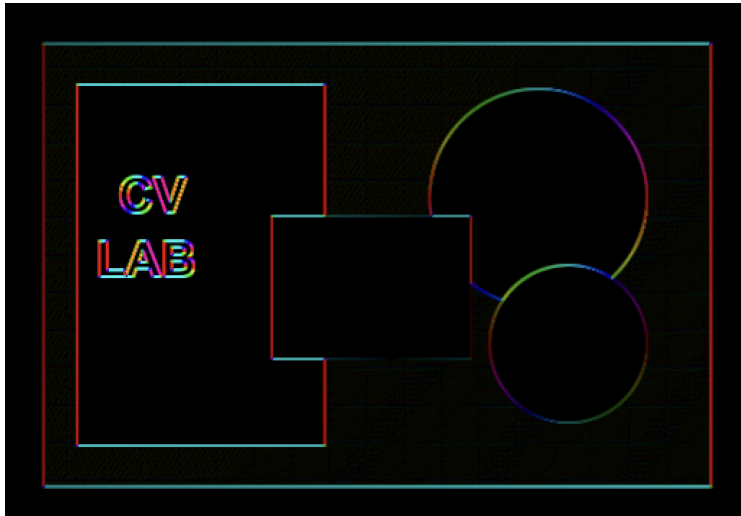
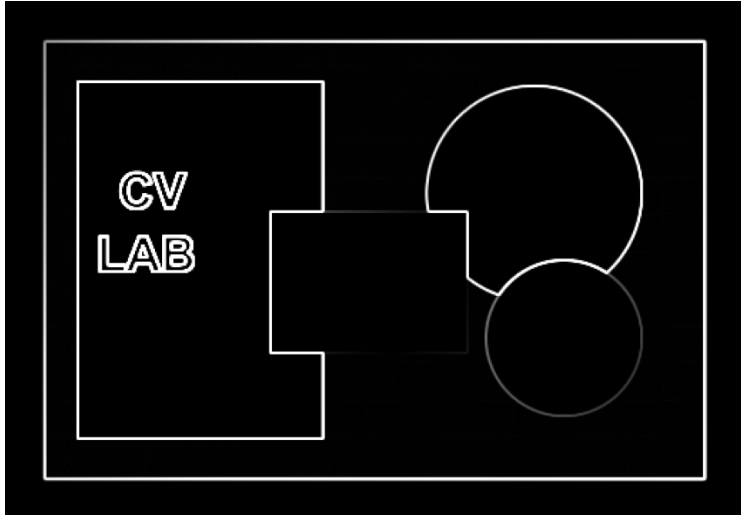
Suppose we increase the Gaussian sigma for smoothing before computing image gradients. What is the most likely effect?

- A. Greater noise sensitivity and more precise localization
- B. Greater noise sensitivity and less precise localization
- C. Less noise sensitivity and more precise localization
- D. Less noise sensitivity and less precise localization

## A1: Smoothing and localization (D)

- A. A larger  $\sigma$  reduces noise sensitivity, not increases it.
- B. The localization part is correct, but noise sensitivity is reversed
- C. A larger  $\sigma$  does reduce noise sensitivity, but it also blurs edges more, so like in option B, edge localization becomes less precise.
- D. Stronger Gaussian smoothing removes more high-frequency noise, making the gradient less sensitive to noise. However, it also spreads an edge over a wider region, making its exact position harder to localize.**

## Q2: Signed filter responses

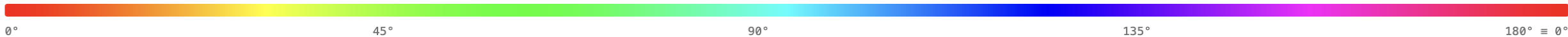
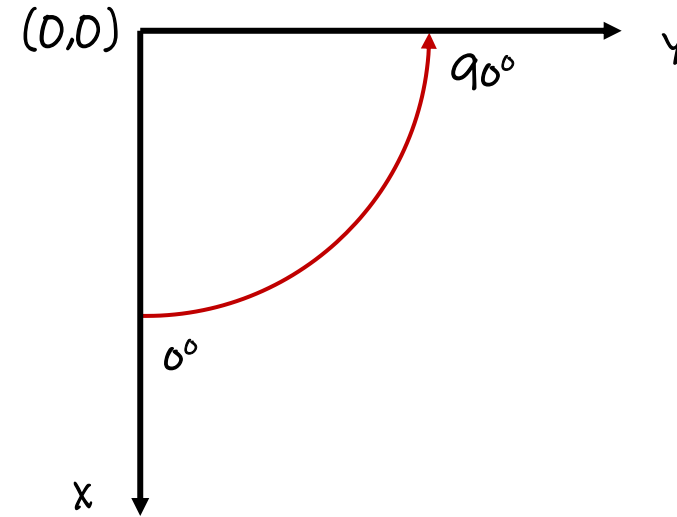


A horizontal derivative filter produces both positive and negative values. Before computing gradient orientation, what should we do?

- A. Retain the signed values.
- B. Set all negative values to zero.
- C. Replace every value by its absolute magnitude.
- D. Rescale the values independently into the range 0–255.

## A2: Signed filter responses (A)

$$\theta = \text{atan2}(G_y, G_x)$$



- A. Keep positive and negative derivative values so the gradient direction can be determined correctly. The direction is determined modulo 180°**

## Q3: Canny Edge

A pixel P has gradient magnitude 80 and a horizontal gradient direction.

Its two neighbouring magnitudes along the gradient direction are 90 and 70.

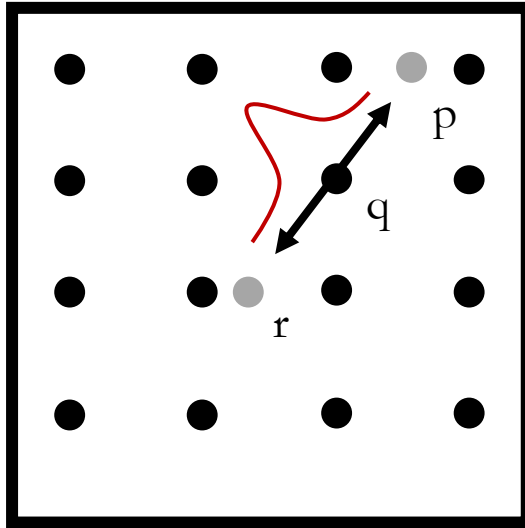
The Canny thresholds are:

- High threshold = 75
- Low threshold = 40

What happens to P?

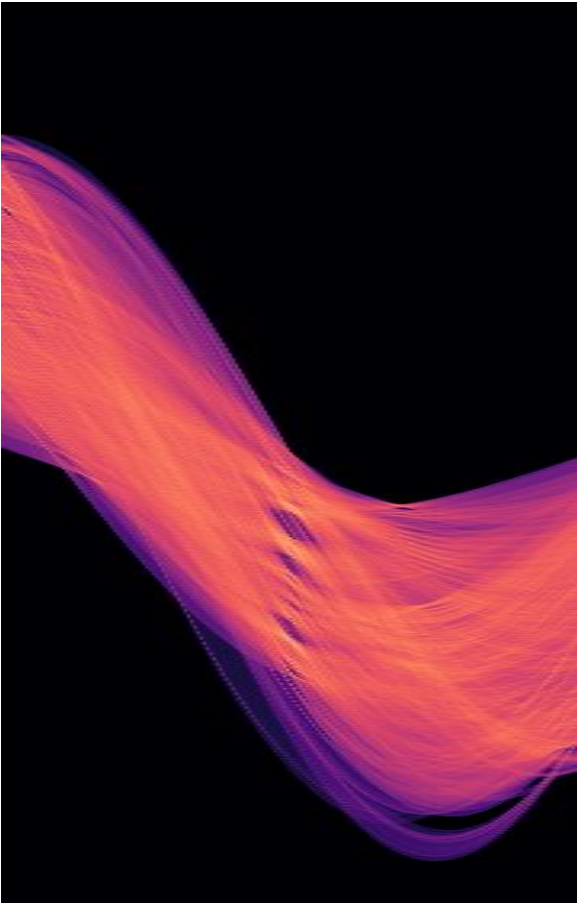
- A. It becomes a strong edge because 80 exceeds the high threshold.
- B. It becomes a weak edge because one neighbouring response is larger.
- C. It is removed during non-maximum suppression.
- D. It is retained only if connected to the neighbouring pixel with magnitude 90.

## A3: Canny Edge (C)



- A. Although  $80 > 75$ , thresholding happens after non-maximum suppression
- B. Similar to A, it is not labeled weak; it is eliminated first
- C. **P is removed because a neighboring response along the gradient direction is larger.**
- D. Hysteresis connectivity is considered only for pixels that survive non-maximum suppression.

## Q4: Hough Transform



In the Hough transform, an image-space point corresponds to:

- A. One set of parameters for a single line
- B. All possible lines passing through that point
- C. One image gradient
- D. One fixed orientation

## A4: Hough Transform (B)

- A. A single image point does not determine one unique line; infinitely many lines can pass through it.
- B. The point votes for all possible lines that could pass through it. In Hough parameter space, these possibilities form a sinusoid.**
- C. An image gradient is a local intensity-change measurement, not what an image point represents in Hough space.
- D. The point is not limited to one line orientation.